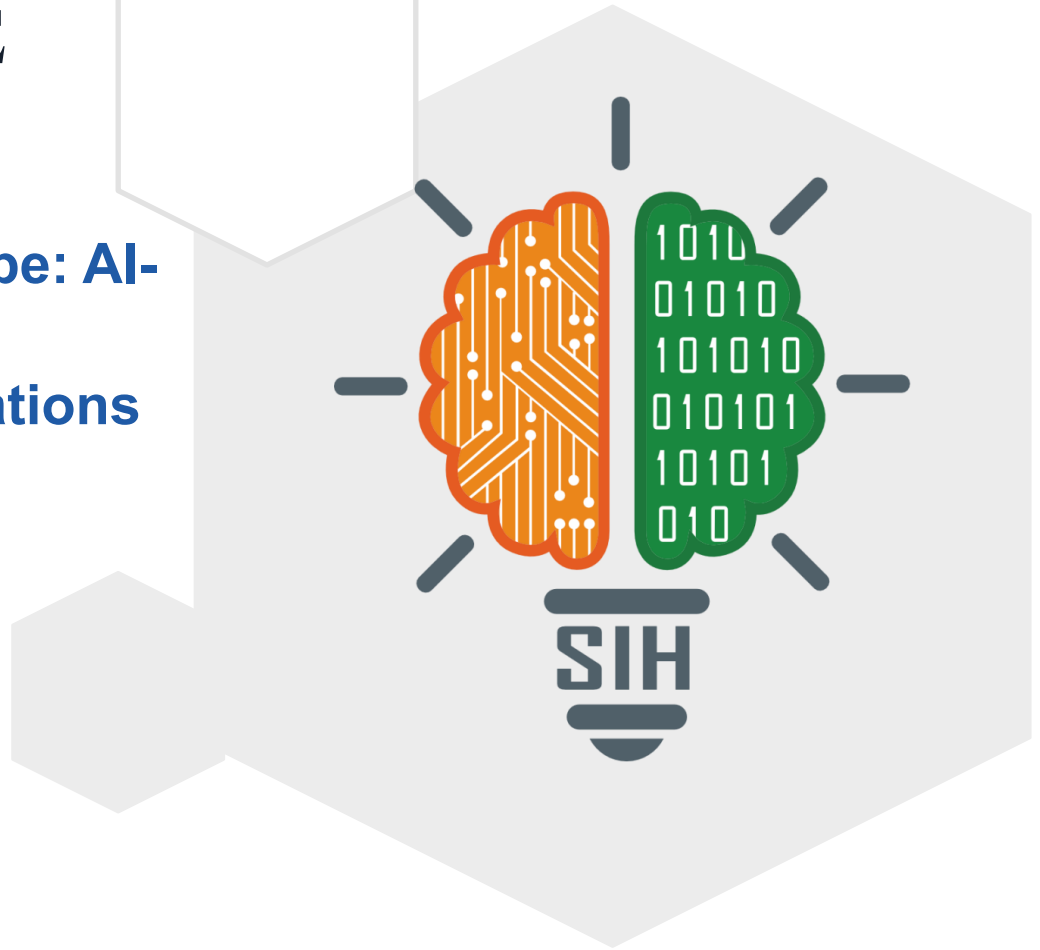
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TITLE PAGE

- **Problem Statement ID –** SIH26159
- **Problem Statement Title –** SecureMailScope: AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications
- **Theme –** Blockchain & Cybersecurity
- **PS Category –** Software
- **Team ID –** [Enter Team ID]
- **Team Name –** [Enter Team Name]



[Team Name]

SecureMailScope

Passive email-security forensics workstation for SMTP · IMAP · POP3 captures

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The Problem

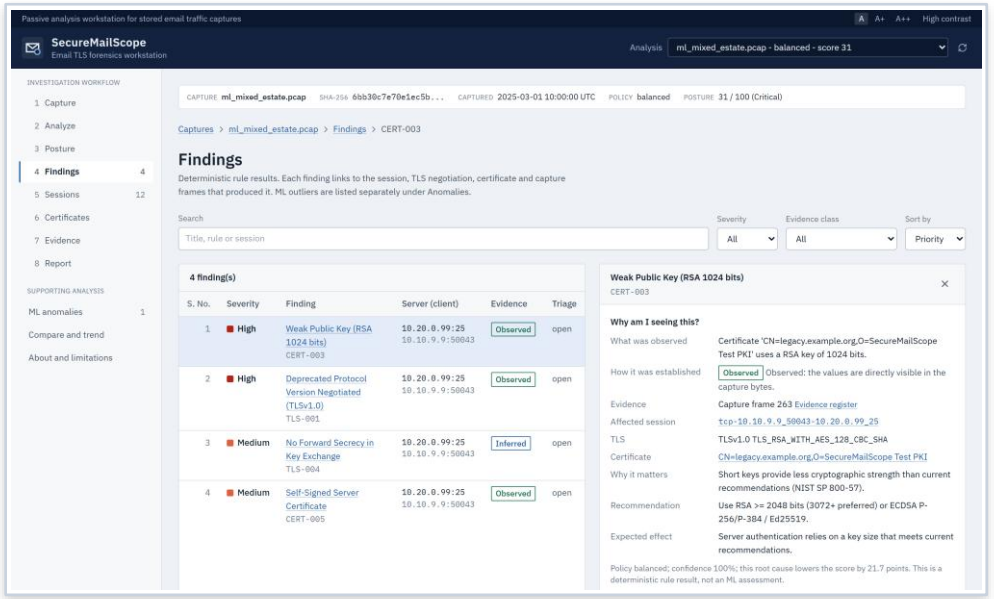
- Mail servers still negotiate **obsolete TLS 1.0/1.1**, **weak ciphers** and **expired or self-signed certificates**.
- **STARTTLS** can be refused or stripped, so mail and passwords travel in **plaintext**.
- Packet tools **decode** traffic but do not **assess posture**; active scanners must **contact live servers** and cannot analyse a capture.

Proposed Solution

- **Upload a PCAP:** TCP streams **reassembled**, **SMTP/IMAP/POP3** identified, **STARTTLS/TLS** and **X.509** analysed.
- **19 deterministic rules** grade each session with an evidence class and the **exact capture frame**.
- **Posture score 0-100** with prioritised remediation; **Isolation Forest** ranks sessions weaker than the capture norm.
- **14-section forensic report** (JSON/HTML/PDF), redacted mode, evidence bundle and **STIX 2.1**.

Uniqueness & Innovation

- **100% passive:** never contacts a mail server and never decrypts traffic.
- **Evidence chain:** every finding traced to session, TLS, certificate and packet frame.
- **Explainable & honest:** 'Why am I seeing this?'; TLS 1.3 certificates marked **NOT OBSERVABLE**.
- **Analyst workflow:** 8 steps, policy profiles, before/after compare, notes, trend.



Working prototype: finding with its evidence and remediation

PCAP → Sessions → TLS + X.509 → Rules → Score → Report

WORKING PROTOTYPE BUILT AND TESTED

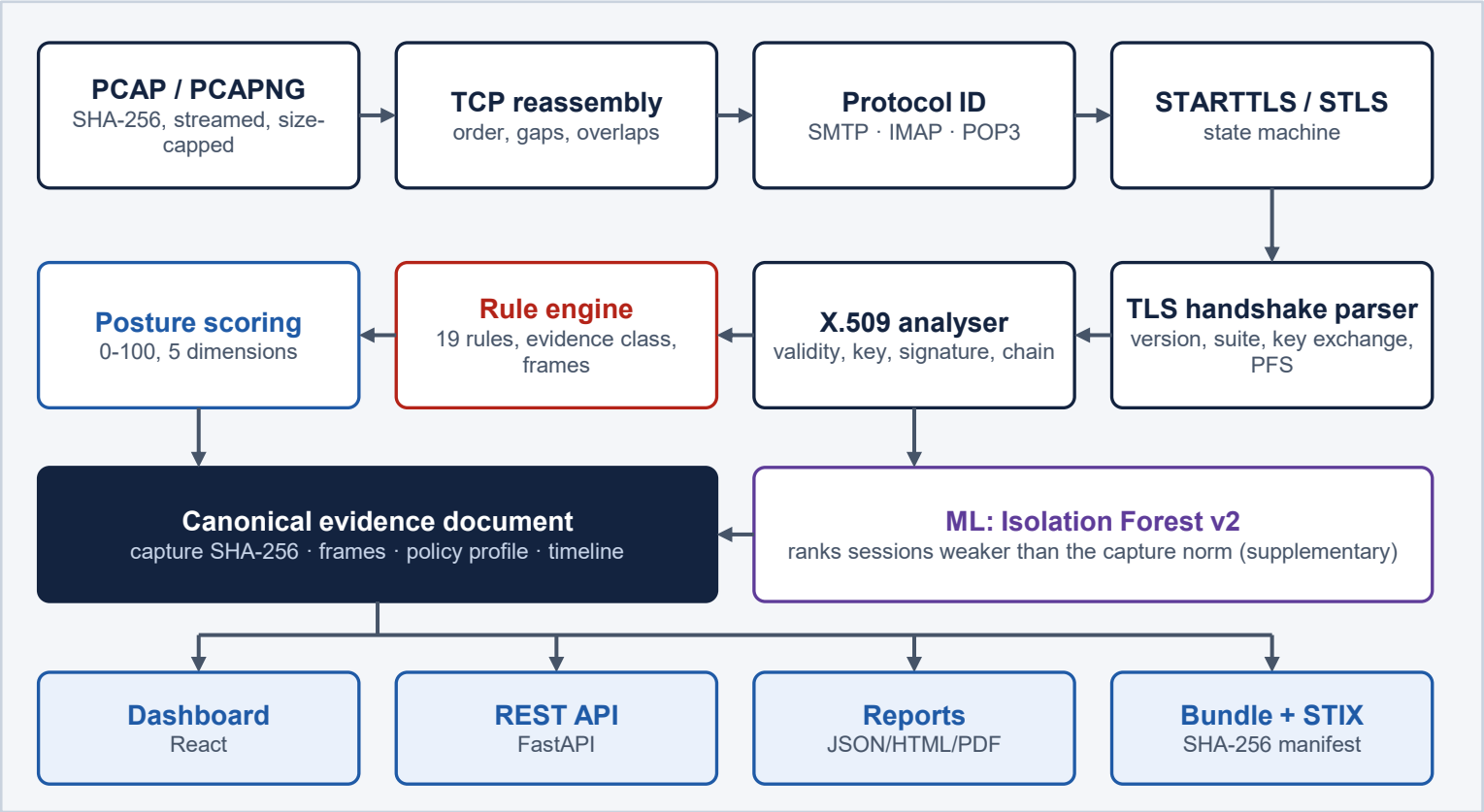
414 automated tests · 112/112 browser checks · 0 accessibility (axe) violations

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TECHNICAL APPROACH

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System Architecture (passive analysis pipeline)



Analyst workflow (8 connected steps)



Languages, Frameworks & Libraries

Python · FastAPI · SQLAlchemy · SQLite

scikit-learn · NumPy · Pydantic · Jinja2

React · TypeScript · Vite · Tailwind

Packet & crypto: dpkt, cryptography (X.509), IANA cipher registry
Reports: ReportLab (PDF), STIX 2.1 (stix2)

Testing & Validation Tools

- **Tests:** pytest, Hypothesis fuzzing, Playwright browser checks
- **Real mail lab:** Mailpit, aiosmtpd, pymap (IMAP) servers
- **Public captures:** Wireshark wiki (Dovecot IMAP/POP3, SMTP)

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FEASIBILITY AND VIABILITY

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Technical Feasibility: built and verified

414
automated tests passing
(unit, fuzz, API, reports)

15 + 5
real lab captures (SMTP, IMAP, POP3)
+ public Wireshark captures

1,000,000
packets (198.7 MB) analysed
end-to-end in about 4 minutes

112 / 112
browser checks on every control;
0 accessibility violations

Potential Challenges and Risks	Strategies for Overcoming These Challenges
TLS 1.3 encrypts server certificates	Report them as NOT OBSERVABLE; analyse TLS 1.2 and earlier certificates; never guess.
Hostile or malformed captures	Bounded parsers and resource limits; 13 Hypothesis fuzz properties; clean errors, no crashes.
No labelled real-world data for ML	Unsupervised, supplementary ranking; evaluated on 80 labelled lab sessions; rules stay primary.
Sensitive traffic (credentials, addresses)	Credentials never stored; redacted outputs; API bound to 127.0.0.1.
Very large captures	Streamed ingestion, 100 MB upload limit, batched ML scoring, capped PDF tables.

Viability

Runs on **one analyst laptop** (Windows / Linux), **offline and air-gapped**; open-source stack with **no licence cost**; deployable by CERT / SOC teams without touching production mail servers.

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IMPACT AND BENEFITS

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IMPACT

Stronger national email security
Government, banks and enterprises find weak TLS before attackers do.

Faster audits
PCAP to prioritised report in minutes: 1M packets in about 4 min on a laptop.

Forensically defensible
Each finding cites capture frames and the capture SHA-256.

Measurable remediation
Before / after compare: same server scored 0, then 100, in our lab.

Privacy-preserving sharing
Redacted reports and STIX 2.1 observations for CERT exchange.

BENEFITS

Zero risk to production
Fully passive: no scanning, no decryption, no traffic generated.

Works offline / air-gapped
Single-laptop deployment for sensitive environments.

Explainable results
'Why am I seeing this?' with evidence, recommendation and effect.

Fits each organisation
Balanced, strict and legacy-audit policy profiles.

Low cost
Open-source stack; no licences or cloud services required.

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COMPARISON WITH EXISTING SOLUTIONS

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	PACKET ANALYSERS (PASSIVE)			ACTIVE TLS SCANNERS		OUR SOLUTION
Capability	**Wireshark**	**Zeek**	**NetworkMiner**	**testssl.sh**	**Qualys SSL Labs**	**SecureMailScope**
Analyses stored PCAP captures offline	✓	✓	✓	✗	✗	✓
Never contacts the mail server (fully passive)	✓	✓	✓	✗	✗	✓
SMTP / IMAP / POP3 incl. STARTTLS	✓	✓	~ 2	✓	✗	✓
TLS parameters and X.509 certificates	✓	✓	✓	✓	✓	✓
Automatic weak-crypto detection (version, cipher, certificate)	✗	~ 1	✗	✓	✓	✓
Email posture score with prioritised remediation	✗	✗	✗	~ 3	~ 4	✓
Finding-to-packet-frame evidence chain	✗	✗	✗	✗	✗	✓
ML ranking of anomalous TLS sessions	✗	✗	✗	✗	✗	✓
Forensic report (JSON/HTML/PDF), redaction, STIX 2.1	✗	✗	✗	~ 3	✗	✓
Before/after comparison and posture trend	✗	✗	✗	✗	✗	✓

✓ Built in ~ Partial / needs manual work ✗ Not provided

1 Zeek: weak-key and expiring-certificate checks through policy scripts that must be configured. 2 NetworkMiner: extracts cleartext mail and credentials; STARTTLS upgrades are not assessed. 3 testssl.sh: per-host rating and JSON/HTML output from an active scan of a live server. 4 SSL Labs: A–F grade for public HTTPS servers only. Based on each tool's public documentation (September 2026); each remains excellent for its own purpose.

Existing tools either **see the traffic** or **grade live servers**. SecureMailScope does both **from a stored capture**, with evidence for every finding.

Why SecureMailScope is better

Two worlds in one tool
Passive like packet analysers, judging posture like TLS scanners.

Evidence you can defend
Finding to session, certificate and packet frame, with capture hash.

Built for email
SMTP, IMAP, POP3 and STARTTLS end to end, incl. downgrade and plaintext login.

Honest, explainable ML
Ranks weaker-than-normal sessions; never claims attacks.

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RESEARCH AND REFERENCES

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Email and TLS standards

- **RFC 3207 - SMTP Service Extension for Secure SMTP over TLS (STARTTLS)**
  https://www.rfc-editor.org/rfc/rfc3207
- **RFC 2595 - Using TLS with IMAP, POP3 and ACAP**
  https://www.rfc-editor.org/rfc/rfc2595
- **RFC 8314 - Cleartext Considered Obsolete: TLS for Email Submission and Access**
  https://www.rfc-editor.org/rfc/rfc8314
- **RFC 8446 - TLS 1.3; RFC 8996 - Deprecating TLS 1.0 and TLS 1.1**
  https://www.rfc-editor.org/rfc/rfc8996
- **RFC 9325 (BCP 195) - Recommendations for Secure Use of TLS**
  https://www.rfc-editor.org/rfc/rfc9325

Cryptographic guidance

- **NIST SP 800-52 Rev. 2 - Guidelines for TLS Implementations**
  https://csrc.nist.gov/pubs/sp/800/52/r2/final
- **NIST SP 800-57 Part 1 Rev. 5 - Key Management (security strengths)**
  https://csrc.nist.gov/pubs/sp/800/57/pt1/r5/final
- **RFC 5280 - X.509 Certificate and CRL Profile**
  https://www.rfc-editor.org/rfc/rfc5280

ML, exchange formats and data

- **Liu, Ting & Zhou - Isolation Forest, IEEE ICDM 2008**
  https://doi.org/10.1109/ICDM.2008.17
- **OASIS STIX Version 2.1 (observations export)**
  https://docs.oasis-open.org/cti/stix/v2.1/stix-v2.1.html
- **Wireshark wiki Sample Captures (external validation)**
  https://wiki.wireshark.org/SampleCaptures

19
deterministic rules

20
real + public validation captures

414
automated tests

14
report sections

Project resources

Source code & documentation
github.com/Kshitij1425/Secure-mail-protocol

- **Problem statement:** SIH26159 (NTRO), SIH 2026
- Methodology, limitations and ML evaluation in the repository documentation